



CHI CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & ROUTER

Installation

```
go get github.com/go-chi/chi/v5
r := chi.NewRouter()
r.Use(middleware.Logger)
r.Use(middleware.Recoverer)
http.ListenAndServe(":3000", r)
```

04. ROUTE GROUPS & MOUNTING

Workflow

```
r.Route("/api", func(r chi.Router) {
  r.Use(JWTMiddleware)
  r.Get("/users", listUsers)
  r.Post("/users", createUser)
})
// Mount sub-router:
r.Mount("/admin", adminRouter())
```

02. ROUTING

Architecture

```
r.Get("/users", listUsers)
r.Post("/users", createUser)
r.Put("/users/{id}", updateUser)
r.Delete("/users/{id}", deleteUser)
r.Patch r.Head r.Options
r.HandleFunc("/path", handler) // net/http c...
```

05. URL PARAMS

CRUD API

```
func getUser(w http.ResponseWriter, r *http.R...
id := chi.URLParam(r, "id")
// or from context:
id := r.Context().Value(chi.RouteCtxKey).(*ch...
w.Header().Set("Content-Type", "application/j...
json.NewEncoder(w).Encode(user)
}
```

03. MIDDLEWARE

YAML / Config

```
r.Use(middleware.RequestID)
r.Use(middleware.RealIP)
r.Use(middleware.Logger)
r.Use(middleware.Recoverer)
r.Use(middleware.Compress(5))
r.Use(middleware.Timeout(60 * time.Second))
r.Use(middleware.Throttle(100))
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. SUB-ROUTERS

Hardening

```
func usersRouter() http.Handler {
    r := chi.NewRouter()
    r.Use(AuthRequired)
    r.Get("/", listUsers)
    r.Post("/", createUser)
    r.Route("/{id}", func(r chi.Router) {
        r.Use(UserCtx)
        r.Get("/", getUser)
        r.Put("/", updateUser)
    })
    return r
}
```

10. COMMON GOTCHAS

Unit Tests

Chi uses {param} not :param style
Wildcard: r.Get("/files/*", serveFiles)
Context values require typed keys to avoid collisions
r.Mount strips path prefix from sub-router
middleware.Logger prints to stdout use custom for structured logs

08. RESTFUL PATTERN

Observability

```
r.Route("/articles/{id}", func(r chi.Router) {
    r.Use(ArticleCtx) // load article from DB
    r.Get("/", getArticle)
    r.Put("/", updateArticle)
    r.Delete("/", deleteArticle)
})
// ArticleCtx middleware sets article in cont...
ctx := context.WithValue(r.Context(), article...
```

09. BUILT-IN MIDDLEWARE

Optimization

```
middleware.RequestID // adds X-Request-Id h...
middleware.RealIP    // sets RemoteAddr to ...
middleware.Logger     // structured request ...
middleware.Recoverer // recover from panics
middleware.Compress(5) // gzip (level 5)
middleware.Timeout(t) // request timeout
middleware.CleanPath  // normalize URL paths
```